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Problems to be discussed:

Comments:

Homework attached on page:

23. If $\tan\theta = \frac{1}{\sqrt{5}}$, then find the value of $\frac{\operatorname{cosec}^2\theta - \sec^2\theta}{\operatorname{cosec}^2\theta + \sec^2\theta}$.

$$\frac{\sin^2\theta\left(\frac{1}{\sin^2\theta} - \frac{1}{\cos^2\theta}\right)}{\sin^2\theta\left(\frac{1}{\sin^2\theta} + \frac{1}{\cos^2\theta}\right)}$$

$$\frac{\sin\theta}{\cos\theta} = \tan\theta$$

$$\frac{a}{b} \times \frac{c}{c} \frac{1 - \tan^2\theta}{1 + \tan^2\theta} =$$

$$\frac{1 - \frac{1}{5}}{1 + \frac{1}{5}} = \frac{4}{6} = \frac{2}{3}$$

$$\frac{ac}{bc} \quad \text{if } c \neq 0$$

$$\sin\theta \neq 0$$

$$\sin\theta = 0 \rightarrow \tan\theta = 0$$

1(B). If $\frac{1}{\sin x - \cos x} = \frac{\operatorname{cosec} x}{\sqrt{2}}$, then prove that $\left(\frac{1}{\sin x + \cos x}\right)^2 = \frac{\sec^2 x}{2}$

$\sin x - \cos x = \sqrt{2}\sin x$

$\sin x + \cos x =$

$\left(\frac{1}{\sin x + \cos x}\right)^2 = \frac{1}{2\cos^2 x}$

$\Rightarrow \boxed{(\sin x + \cos x)^2 = 2\cos^2 x}$

$\sin x + \cos x = -(\sqrt{2})\cos x$

$\boxed{(1-\sqrt{2})\sin x = \cos x}$

$+\sin x + \sin x$

$(2-\sqrt{2})\sin x$ ✓

$(2-\sqrt{2}) \times \frac{\cos x}{1-\sqrt{2}}$ ✓

$\frac{(2-\sqrt{2})(1+\sqrt{2})}{(1-\sqrt{2})(1+\sqrt{2})}\cos x$

12. The mean age of a combined group of men and women is 35 years. If the mean ages of the men and women are 38 years and 30 years respectively, then the percentage of women in the group is: 1

(A) 15 (B) 25.5 (C) 35 (D) 37.5 ✓

Sum of the ages of men → A_m
" women → A_w

$$\frac{A_m + A_w}{m+w} = 35$$

$$\frac{A_m}{m} = 38 \quad \& \quad \frac{A_w}{w} = 30$$

$A_m = 38m$, $A_w = 30w$

$$\frac{38m + 30w}{m+w} = 35$$

$38m + 30w = 35m + 35w$

$3m = 5w$

$\frac{m}{w} = \frac{5}{3}$

$\left(\frac{w}{m+w} \times 100\right)$

$\left(\frac{1}{\frac{m}{w}+1} \times 100\right)$

$\frac{1}{\frac{5}{3}+1} \times 100 \rightarrow \frac{3}{8} \times 100 = 37.5$

37.5

21. If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + (b^2 - ac) = 0$ are real and equal, show that either $a = 0$ or $(a^3 + b^3 + c^3) = 3abc$. [CBSE 2017]

$$[2(a^2-bc)]^2 - 4(c^2-ab)(b^2-ac) = 0$$

$$4[a^4 - 2a^2bc + \cancel{b^2c^2} - \cancel{b^2c^2} + ac^3 + ab^3 - a^2bc] = 0$$

$$\hookrightarrow a^4 + ac^3 + ab^3 - 3a^2bc = 0$$

$$a(a^3 + c^3 + b^3 - 3abc) = 0$$

22. If the roots of the equation $x^2 + 2cx + ab = 0$ are real and unequal, prove that the equation $x^2 - 2(a + b)x + a^2 + b^2 + 2c^2 = 0$ has no real roots.

$D > 0$

$\Rightarrow (2c)^2 - 4ab > 0$

$4c^2 - 4ab > 0$

$c^2 - ab > 0$

D'

$\Rightarrow 4[(a+b)^2 - (a^2+b^2+2c^2)]$

$\Rightarrow 4[2ab - 2c^2]$

$\Rightarrow 8(ab - c^2)$

but $ab - c^2 < 0$

$\Rightarrow D' < 0 \Rightarrow$ No real roots



(ii) The sums of the first m and n terms of an A.P. are in the ratio $(2m + 1) : (3n + 1)$. Find the ratio of its 7th and 10th terms.

[Wrong]

If the ratio of the sums of n terms of two APs is $(7n + 1) : (4n + 27)$, find the ratio of their (i) mth terms and (ii) 9th terms.

$$\frac{a+(m-1)d}{A+(n-1)D} = \frac{4m-6}{8m+23}$$

$$\frac{\frac{n}{2}[2a+(n-1)d]}{\frac{n}{2}[2A+(n-1)D]} = \frac{7n+1}{4n+27}$$

$$\frac{a+\frac{(n-1)}{2}d}{A+\frac{(n-1)}{2}D} = \frac{7n+1}{4n+27}$$

$$\frac{n-1}{2} = m-1$$

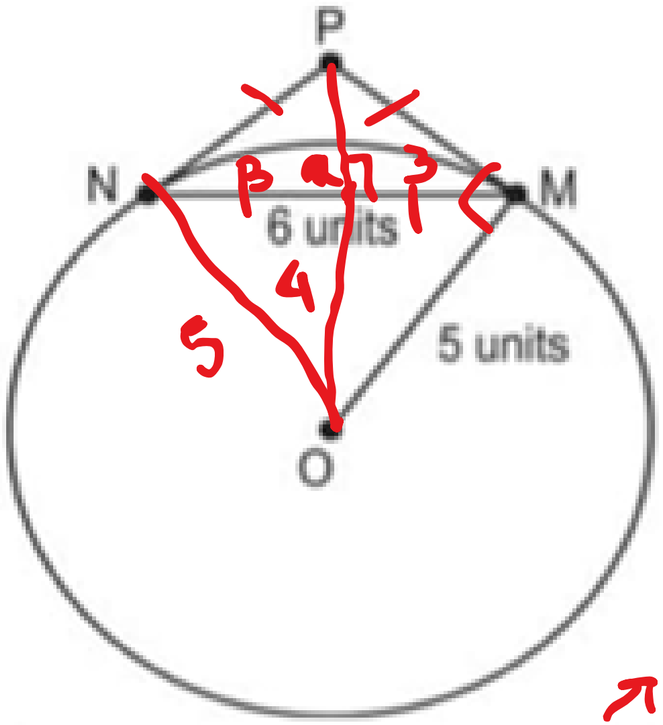
$$\Rightarrow n-1 = 2m-2$$

$$n = 2m-1$$

$n = 2m-1$

$$= \frac{14m-6}{8m+23}$$

16. In the given figure, there is a circle with centre O and radius 5 units. PM and PN are tangents and length of the chord MN is 6 units.



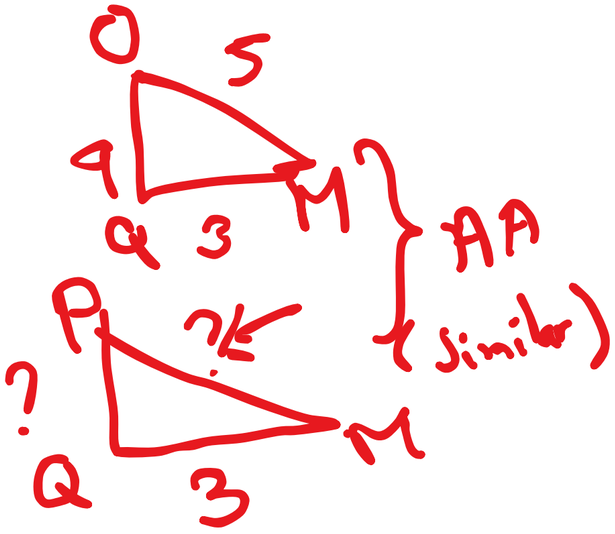
Find the length of PM + PN. [Note : The figure is not to scale]

MQ = NQ = 3u

OQ = ? 4u

in △PQM

in OMQ ~ PMQ



$\frac{15}{2} = 7.5u$

$\angle P = \angle M$

$\triangle OQM \sim \triangle MQP$

$\frac{OM}{PM} = \frac{4}{3} = \frac{OQ}{QM}$

$PM = \frac{15}{4}$

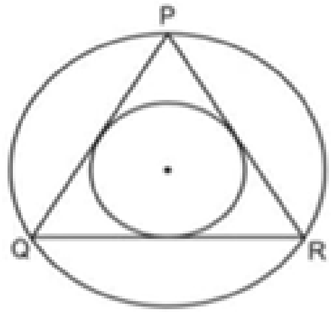
$\pi(T^2 - t^2)$

$= \pi \frac{a^2}{4}$

$A = \frac{a^2}{\sqrt{3}}$

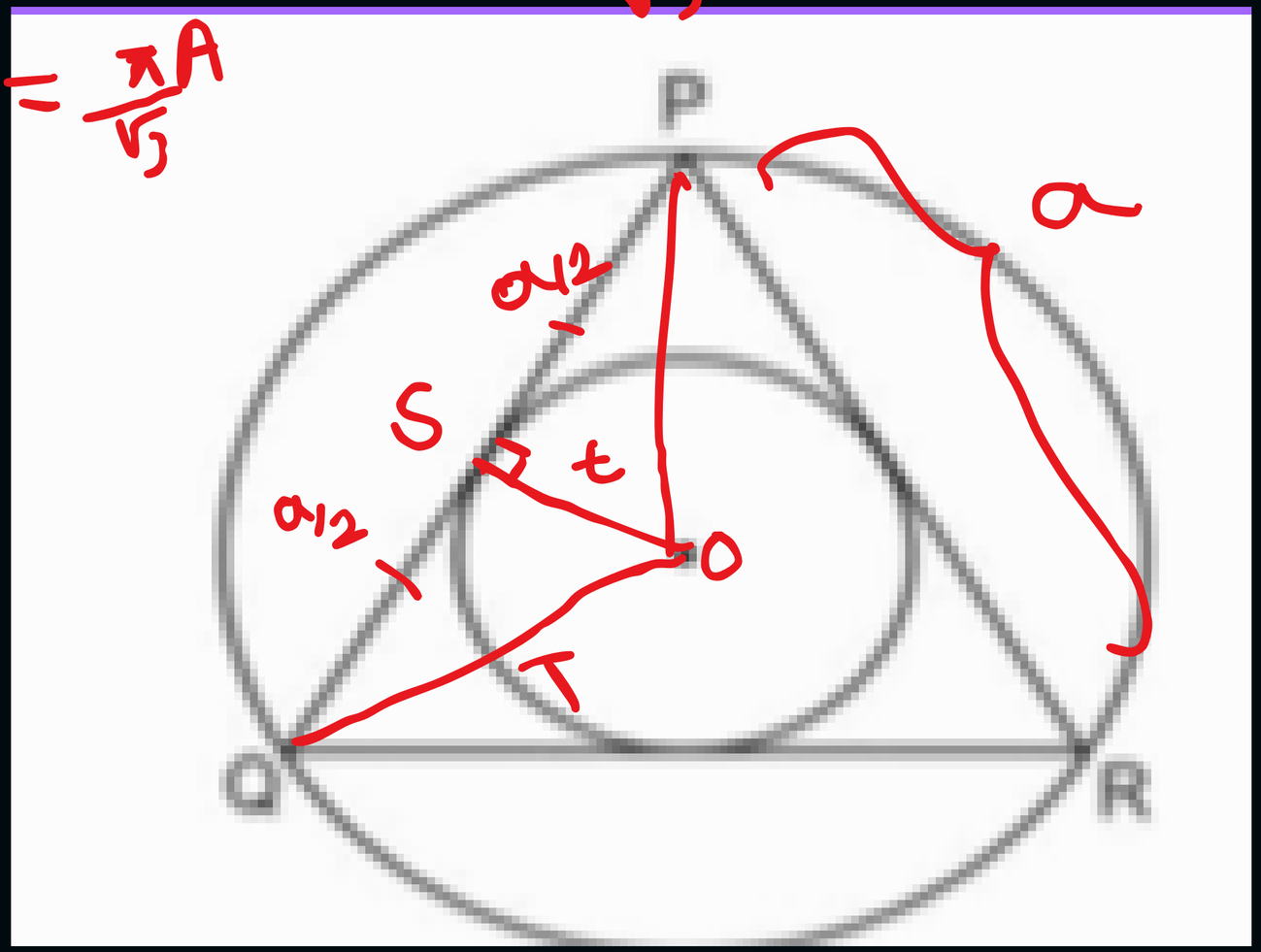
$= \frac{\pi A}{\sqrt{3}}$

17. In the figure given below, O is the centre of two concentric circles. □ PQR is an equilateral triangle such that its vertices and sides touch the bigger and smaller circles respectively. The difference between the area of the bigger circle and the smaller circle is 616 cm^2.



(Note: The figure is not to scale.)

Find the perimeter of □ PQR. (Note : Take $\pi = \frac{22}{7}$)



$T^2 - t^2 = 196$

$\frac{a^2}{4} + t^2 = T^2$

$\frac{a^2}{4} = 196$

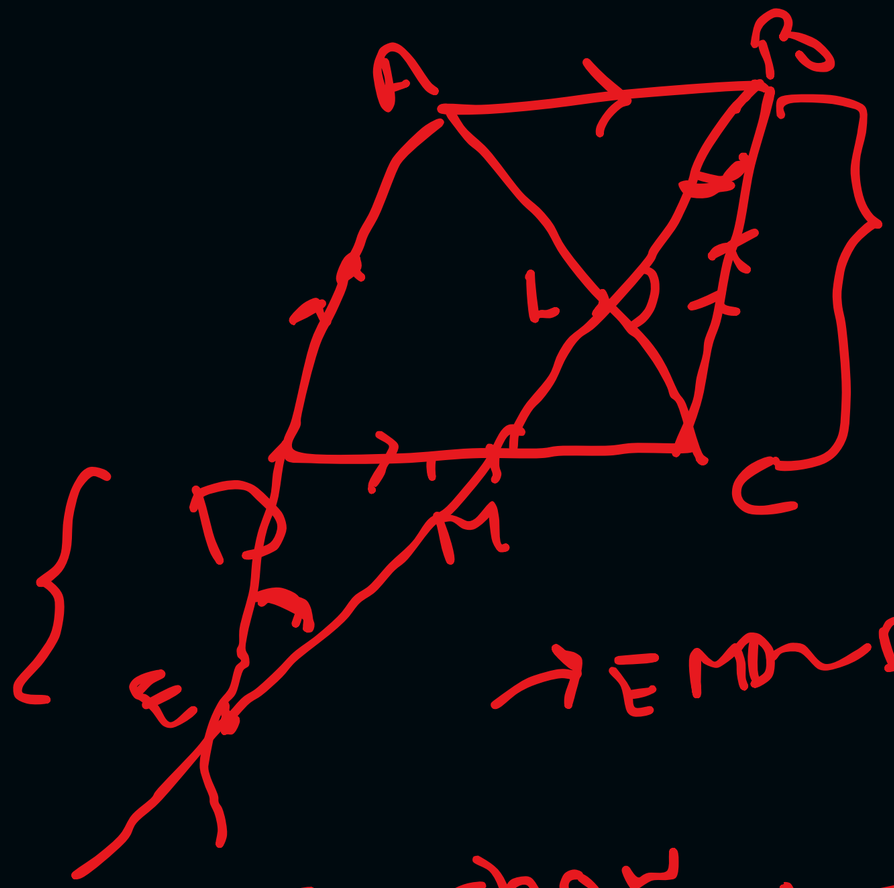
$\frac{a}{2} = 14$

$28 \times 3 = 84$

$a = 28$

2

EXAMPLE 41 *Through the mid-point M of the side CD of a parallelogram ABCD, the line BM is drawn intersecting AC in L and AD produced in E. Prove that EL = 2BL.* [CBSE 2009]



$\frac{EL}{BL} = 2$

$\sim ALE$ $= \frac{AL}{CL}$

BL

$\rightarrow EMD \sim BMC$ $\quad ALE \sim CLB$ $= \frac{AE}{CB}$

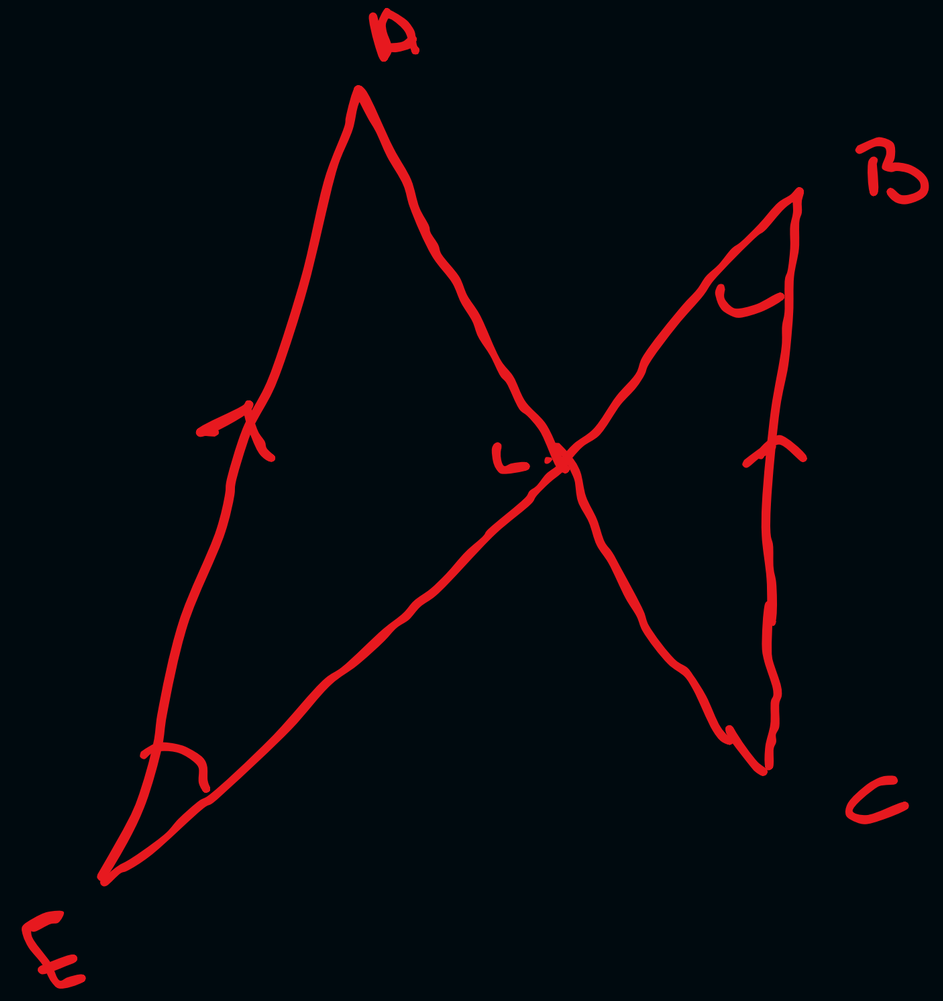
We show $DE = BC$

$= \frac{AD + DE}{BC}$

$= \frac{BC + DE}{BC}$

We show To

$\frac{DE}{BC} = \frac{DM}{MC} = 1$

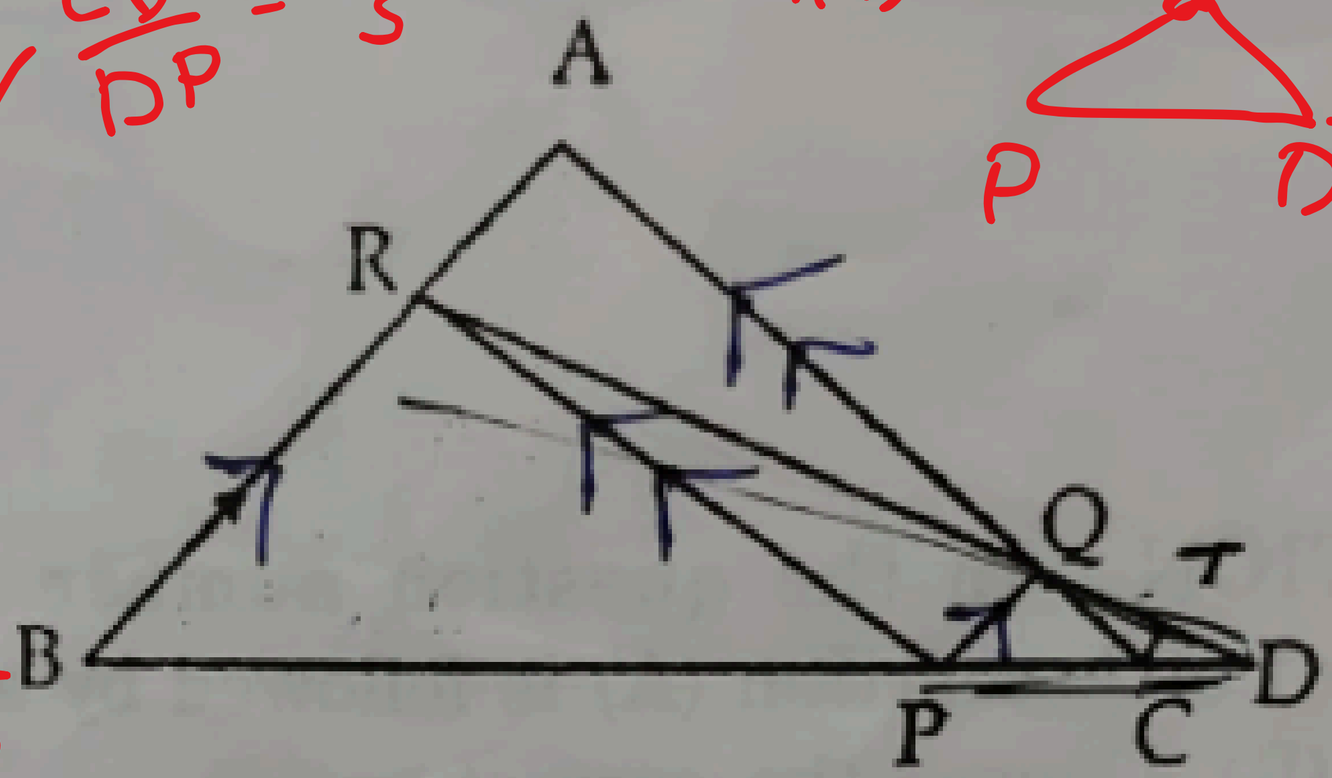


$\angle ALE = \angle BLC$

$\angle AEL = \angle LBC$

$\triangle ALE \sim \triangle CLB$

22. P is any point on side BC of ΔABC, $PQ \parallel BA$ and $PR \parallel CA$. The line QR when extended meets CB at D. Prove that $DP^2 = BD \times CD$. Also find $BD \times CD$, if PD = 12 cm.



$$\frac{1}{5} = \frac{DP}{BD} = \frac{CD}{DP} = \frac{t}{s}$$

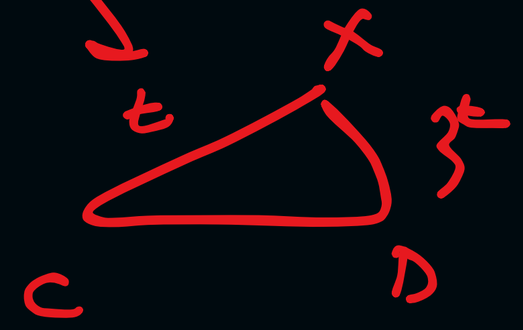
$$\frac{CX}{P} = \frac{DZ}{C}$$

$$\frac{CX}{ZP} = \frac{DZ}{X}$$

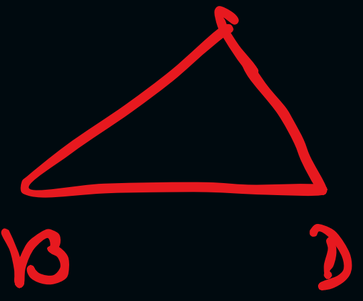
$$X_D = DZ$$

$$X \cdot D = ZP$$



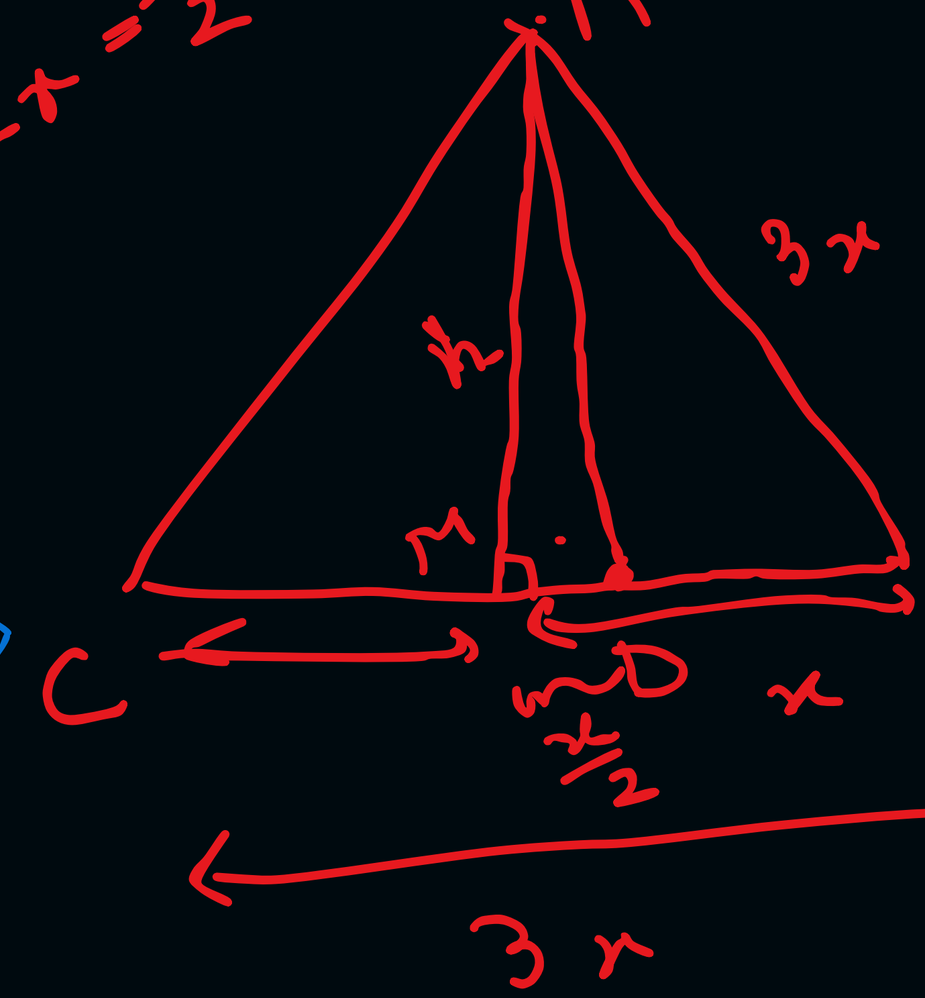


2





23. In an equilateral ΔABC, D is a point on side BC such that $BD = \frac{1}{3}BC$. Prove that $9(AD)^2 = 7(AB)^2$.



$BM = MC = \frac{3x}{2}$

$DM = \frac{3x}{2} - x = \frac{x}{2}$

$\Delta ADM \Rightarrow AD^2 = h^2 + \frac{x^2}{4}$

$= \frac{28x^2}{4} = 7x^2$

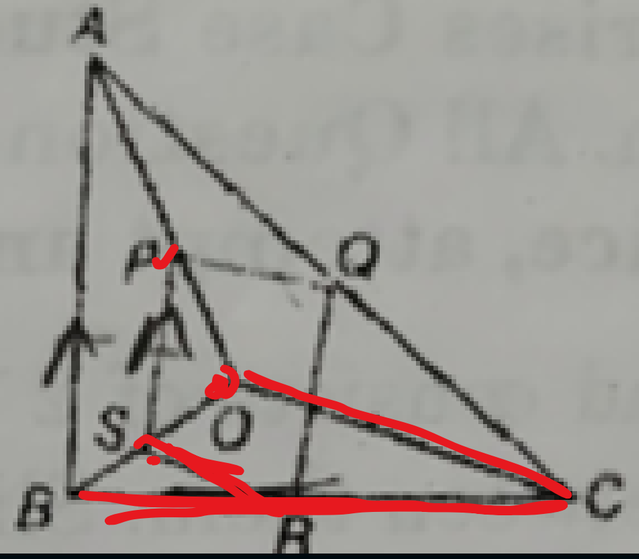
$\Delta ABM \Rightarrow 9x^2 = \frac{9x^2}{4} + h^2$

$\Rightarrow h^2 = \frac{27}{4}x^2$

$\frac{AD^2}{AB^2} = \frac{7x^2}{9x^2}$

$\Rightarrow 9AD^2 = 7AB^2$

34. In the given figure, if PQRS is a parallelogram and AB||PS, prove that OC||SR. [5]



$\triangle OPS \sim \triangle OAB$

$\frac{OS}{OB} = \frac{PS}{AB}$

$\triangle CQR \sim \triangle CAB$

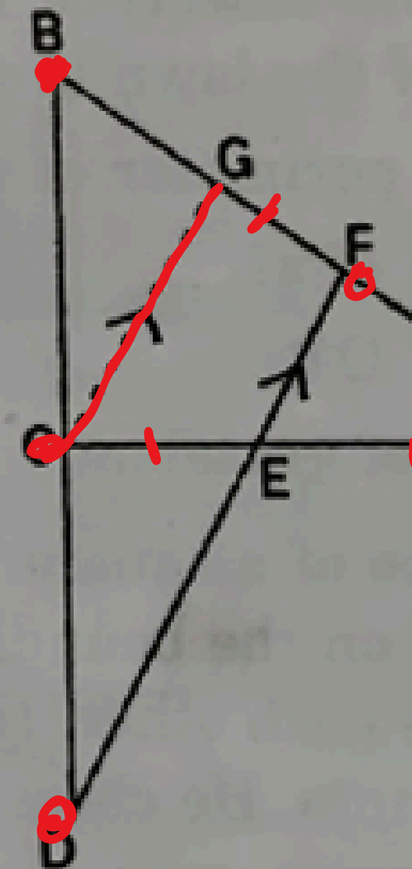
$\frac{CR}{CB} = \frac{PS}{AB} = \frac{QR}{AB}$

$\frac{OS}{OB} = \frac{CR}{CB} \Rightarrow \frac{BS}{OB} = \frac{BR}{BC}$

[$\triangle OBC$, BPT]

34. In the given figure, line segment DF intersect the side AC of a triangle ABC at the point E such that E is the mid-point of CA and $\angle AEF = \angle AFE$.

Prove that $\frac{BD}{CD} = \frac{BF}{CE}$



$\frac{BD}{CD} = \frac{BE}{CE}$

$\triangle BCG \sim \triangle BDF$

$\frac{BC}{BD} = \frac{BG}{BF}$

$1 - \frac{BC}{BD} = 1 - \frac{BG}{BF}$

$= \frac{CD}{BD} = \frac{GF}{BF} = \frac{CE}{BF}$